Measuring Safety, Not Benchmark Scores: Infrastructural Barriers to Mental-Health AI Evaluation

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Large language models are increasingly used for mental-health support, yet the field still lacks reliable methods for determining whether these systems are safe for vulnerable users. This paper examines the barriers preventing mental-health AI evaluation from producing evidence that is transparent and useful for governance. We draw on interviews with 20 benchmark developers, clinicians, lived-experience advocates, and policy researchers to identify recurring challenges in the design, maintenance, and use of mental-health safety evaluations. Across interviews, we find that progress is constrained by misaligned incentives, siloed evaluation tools, limited access to real-world interaction data, and uncertainty about how existing benchmark results should inform regulation. We also find that mental health is inherently pluralistic: definitions of harm vary across populations and clinical perspectives, making a universal safety standard both unrealistic and undesirable. We argue that the field should prioritize building a shared evaluation infrastructure with transparent comparative frameworks and sustained interdisciplinary collaboration so that different definitions of safety can be meaningfully compared rather than reduced to isolated benchmark scores.

1. Intro and Motivation

In the US, over 60 million adults experienced a mental illness last year. Meanwhile, 137 million people live in areas with a shortage of mental-health support, where only 25% of the demand in therapists is met. This is all not to mention the cost that comes with seeing a therapist that can be entirely unaffordable, or the enduring stigma associated with it in many parts of the country and the world. Millions of users in such situations are now turning to AI for conversations that they cannot have with a human. A large language model (LLM) is available at all hours at the touch of a button, costs next to nothing, and gives judgment-free human-like responses. AI has also shown signs of being effective: in the first randomized trial of a generative-AI therapy chatbot, participants with depression saw a 51% average reduction in symptoms and rated the service as comparable to a human therapist. [SAMHSA 2024 NSDUH; HRSA Designated HPSA Quarterly Summary, Dec 2025; Heinz et al., NEJM AI 2025]

However, if AI is going to close the therapy access gap, we must be able to measure whether it is safe for use, to what extent, and for which populations, which turns out to be a Herculean task. The axis of mental health is a relatively unexplored dimension of AI safety research, which is usually positioned along the lines of national security or personal data security. While many AI safety researchers focus on the risks of AI exposing the security vulnerabilities of governments and mega-organizations, few are thinking about the risks of AI exposing and exploiting the vulnerabilities of individuals.

Despite the limited public attention, mental health safety research has drawn the interest of benchmark developers, clinicians, lived-experience advocates, and policy researchers: experts in their respective disciplines. We wanted to better understand the roadblocks they were facing in making progress towards psychologically safer AI. To do this, we interviewed 20 experts, asking questions towards one primary goal: to uncover the challenges experienced in building

. Manuscript submitted to ACM

AI evaluation tools for mental health. We found that there was surprisingly little transparency across the groups of people studied, but that they had overlapping challenges. In this paper, we examine the primary hurdles we uncovered.

<Related Works Section placeholder>

2. Methodology

2.1 Participants and Interviews

We conducted semi-structured interviews with 20 experts with experience working at the intersection of AI evaluation and mental health. Our participants included mental health benchmark developers, clinicians and policy researchers. Our main inclusion criteria for the participant group was that they should have developed, studied, used, or been affected by mental-health AI evaluation practices. We recruited the participants via email, and most participants had an existing professional relationship with one of the authors.

The primary research question that guided the interviews was: What challenges do experts encounter in building and using AI evaluation tools for mental health?

We used semi-structured interviews so that the participants could identify challenges based on their own experiences, while enabling the interviewer to dive into details specific to the participant's experiences. The interviews broadly covered participants' experiences with mental health benchmark development, sources of data used (if any), interactions with other experts in the AI/ mental health domain, and thoughts on regulation. Each interview was approximately 60 minutes long and was conducted remotely between July 1, 2026 and September 1, 2026. With the consent of the participants, we recorded the interviews and took notes. To protect the privacy of the participants, we report their professional role but omit their name and institutional affiliation. The list of participants and their IDs are found in the table below.

Table 1:

Mental Health Experts	Mental Health Benchmark Developers	AI Policy Experts
Clinician (n=4) HCI mental health expert (n=2) Cognitive Science expert (n=1)	Benchmark Developer (n=5) AI Evaluation Researcher (n=1) Clinical AI Developer (n=2) AI Research Engineer (n=1)	Policy Researcher (n=3) AI Policy Advisor (n=1)
P2, P3, P8, P9, P16, P18, P17	P4, P7, P11, P12, P13, P14, P15, P19, P20	P1, P5, P6, P10

2.2 Analysis

We conducted a reflexive thematic analysis [Braun and Clarke], and used inductive reasoning similar to that used by Kapania et al. [cite] The participant interviews were used as evidence from which patterns were identified and organized into higher level themes.

The analysis proceeded in several stages. First, all the interview transcripts were cleaned and manually checked for fidelity, and each participant was assigned a participant ID. Then, the authors read the transcripts and tagged all statements relevant to our research question. These first-level codes included participant pain points, key experiences, and needs, and were all specific to the participant's own interviews. In total, 322 codes were created.

Once all transcripts were tagged accordingly, all first-level codes were exported into a spreadsheet, with notable fields being the participant ID, the code itself, the cited evidence, and a quote ID. This was done to simplify the process of searching for evidence later in the process. These codes, along with the participant and quote IDs were then exported onto a virtual whiteboard for visual collaboration. Here, we compared codes across interviews and noted any recurring patterns and relationships. These codes were manually grouped into broader analytical categories, while maintaining the original code as supporting evidence. For example, a code about a lack of human-AI interaction data from frontier labs could be placed next to a code about the nuances of working with lived-experience groups for grounded data. Both of these support a larger picture of data access issues.

During this project, members of the research team met to collectively group the identified codes and disambiguate boundaries of analytical buckets with one another. These discussions helped clarify groupings and we often referred to the spreadsheet of evidence from the interviews to help us decide where to place certain codes.

Finally, we were able to group codes containing recurring challenges, observations, and needs into three broader themes: incentive misalignment across groups, access to data and its implications to regulation, and naturally occurring pluralism. We use anecdotal evidence from participant interviews to support these themes in the following sections.

2.3 Limitations

The themes centered in this paper do not constitute an exhaustive list of characteristics of challenges in the mental-health AI evaluation domain, but an interpretive account of barriers that were described by participants.

3. Findings and Implications

At a high level, we found that the bottleneck in AI mental health safety evaluation stems from a shortage of a shared methodology and validated infrastructure. There is incentive misalignment and lack of access to data that can provide evidence of harms, which also leads to regulatory struggles. And there is no framework that makes room for multiple perspectives to co-exist. Until the field prioritizes allocating resources to that infrastructure over rewarding leaderboard-style scores, the safety metrics produced by benchmarks will only offer an incomplete view of human-LLM interactions in the mental health domain. [Thesis] We examine the three major themes uncovered below.

While opposing voices might say that this is nothing new, and that all subfields of AI evaluation face pluralistic voices and misaligned incentives, the sensitive and often stigmatized nature of mental health and the high risk for negative human outcomes makes this an especially critical domain to improve.

4. Misaligned Incentives

4.1 Tool Maintenance

Across conversations, a theme that stood out clearly was incentive misalignment. First, the field knows what is methodologically lacking, yet no one is well-incentivized to build the tools and maintain the infrastructure. Developing a novel benchmark draws credit and attention in academia, not spending time maintaining it months later. One researcher, who developed a Python evaluation framework for frontier AI models, described sustaining it as essentially unpaid labor, and a policy researcher pointed to a graveyard of grant-funded tools that died as soon as grant funding ended. This begs the question of whether ephemeral novelty is being unfairly rewarded over consistency and growth over time.

4.2 Safety Scores

There are also misaligned incentives for producing safety scores among industry labs that develop safety benchmarks for the same use cases. One expert on digital mental health pointed out that in a market where companies compete for market share, benchmark developers are quite averse to adopting a competitor's instruments, so collaboration is usually out of the question. This leads to echo chambers where benchmark and testing infrastructure is developed in a silo, and labs become incentivized to "benchmaxx," or optimize models to score highly on benchmarks rather than improving their actual capabilities. Here, developers report high safety scores on models that have been trained to perform well on internally built benchmarks. These safety scores are merely marketing signals that their product performs better than a competitor's product, but there is no grounding for what the score actually represents in practice. Benchmark developers in industry are incentivized to do what is best for the business, not for the public.

4.3 Data (In)Access

Next is the problem of data incentives. The human-AI interaction data that has the power to make any evaluation ecologically valid is also extremely sensitive. Besides the privacy issues of releasing real-users' vulnerable and personally identifiable conversation data, frontier AI labs are not incentivized to invite any sort of scrutiny to the safety and efficacy of their models. As a result, academic researchers do not get access to this data, and the frontier lab researchers that do use this data are often incentivized to minimize any negative findings.

To combat the data access barriers, some researchers resort to creating plausible interaction data (or "user simulation" data) with input from people with first-hand experience of mental health conditions. This is done through working with lived-experience advocacy groups. In our conversations, we learned that many researchers themselves do not approach these groups with the goal to maintain sustainable relationships. They often come with the agenda of collecting data for their research work and disappear once they have made personal gains, without accrediting the right organizations and often without paying them.

As a result, frontline organizations are increasingly averse to sharing data with any researchers. Researchers themselves are not well-incentivized to create a sustainable pipeline from field data to their final output.

One computer science expert and professor described this landscape as one in which "everyone wants to be the center": there are dozens of benchmarks, each locally rational to produce, with none of them accumulating into shared knowledge. Funders, venues, and companies must treat a shared infrastructure as the deliverable, otherwise the field will keep producing "precise" measurements that have no impact.

5. Regulation

5.1 Timing

Policymakers also need convincing evidence that intervention is warranted. Without concrete data that gives evidence of the development of harms, it is difficult for policymakers to push AI safety policy to protect vulnerable users. However, delaying regulation until a corpus of evidence is documented could take years and harms will accumulate in the meanwhile. This was the case with social media regulation.

The researchers we interviewed all analogized harms from the expanding AI companion space to harms that came from unregulated social media (cite: Omegle, facebook/instagram predatory and bullying behavior), which was deployed before the adverse effects were fully understood. By the time governments began responding, it was already deeply

embedded into people's lives. Recently, fines have caught up to big tech companies (Meta) as a result of social media causing online exploitation, addiction, and mental health issues in the youth that grew up using it.

Regulating too late allows the accumulation of preventable harms, but regulating too early can also create problems. If compliance requirements are poorly designed, companies may simply optimize around the tests. Overly restrictive regulation could also remove tools that some people genuinely find beneficial, particularly people in areas with limited access to mental-health services. It is clear that regulation is needed, the question is when and how to do so effectively.

5.2 Global Regulation

Many of the people who benefit from AI tools for mental health reside beyond national lines. AI could widen access to therapy resources to the global population, most of which have no access to mental health resources [cite]. However, from our interviews, we learned that many lawmakers in Europe and Asia do not see mental health as a top-of-mind issue, irrespective of its perceived relevance to AI safety. One researcher in Asia who develops user simulations for mental health was not able to find a clinician capable of advising her work due to the lack of resource allocation to mental health. In interviewing two policy researchers based in Europe or familiar with European policy, we learned that mental health is seen as too progressive a construct. Many with mental health conditions are often stigmatized as features of ill human behavior that do not need to be explicitly protected [cite]. In a world where the Brussels Effect normally takes place (effect where EU regulations get adopted beyond the EU), the US will have to forge their own path to regulation.

The American system is further ahead, but policy researchers need evidence to convince lawmakers that change is needed. Ideally, this would be in the form of real-world human-AI interaction data that shows proven negative consequences. Due to the data access issues discussed above, this is not possible. There is a growing documentation of harms caused by LLMs, but this largely remains unverified by real data.

5.3 Clarity Challenges

The challenge extends beyond convincing policymakers that there is a problem and that they need to act on it now. Lawmakers need guidance towards what constitutes an unsafe system, what evidence concretely demonstrates harm, and what level of risk should trigger intervention. In mental health, those questions are unusually difficult to answer: a response one clinician considers appropriately cautious may be seen by another as unnecessarily restrictive. Even if policymakers agree that psychological safety deserves protection, they still face a more fundamental question: whose definition of psychological safety should be encoded in regulation?

This creates some tension: prematurely encoding one definition of safe behavior into regulation risks putting one particular clinical framework or notion of harm on a pedestal before the field has matured. The regulatory problem therefore cannot be separated from the measurement problem.

6. Pluralism

Mental health constructs are multidimensional and context-dependent, and this domain has always felt the presence of plurality in competing clinical definitions and viewpoints. As a result, it is tempting but unrealistic to immediately call for a single standard of LLM evaluation. The interviews we conducted confirm that plurality is normal and expected. One researcher who studies machine and human cognition referred to therapy-work as epistemically subjective, making it difficult for clinicians to even come to a shared conclusion on the manifestation of certain conditions, or what the appropriate intervention should be.

A professor of psychiatry made the broader point that clinical psychology itself has not resolved many of the questions that AI researchers now implicitly want benchmarks to resolve. We therefore should not expect benchmark designers to produce one universally accepted version of concepts such as dependency, overreliance, or "harmful" advice. An HCI professor gave the further insight that pluralism is not limited to clinical definitions of harm, but also in how a help-seeker may frame their distress linguistically: across many cultures it is the norm to use euphemisms or idioms to describe a health condition. An evaluation designed for a very different population may fail to recognize culturally specific language as help-seeking, and may miss any accumulation of harms.

The plurality in existing evaluation research is natural and desirable in this domain: different benchmarks may represent different populations, clinical traditions, cultural contexts, or harms. The existence of multiple benchmarks only becomes a problem when their results are uninterpretable relative to each other.

7. A Path Forward/ Discussion (include limitations) - Look up positionality and include maybe?

The problems described above do not point toward the need for one definitive mental-health safety benchmark. Instead, they argue for the creation of and investment in building a reliable technical infrastructure and a collaborative network of experts for AI evaluation.

First, researchers should prioritize the creation of transparent evaluation protocols that make clear what a benchmark actually measures. Safety scores should specify the populations they represent, the types of harm they target, and the assumptions under which their results are valid. At the same time, evaluation environments should be designed with pluralism in mind, with the goal to compare multiple evaluation methods against shared definitions, user simulations, or cultural contexts. For this to happen productively, there needs to be a shift in the reward structure. Funders and publishing venues can support this effort by treating the development and maintenance of such tooling as a creditable research contribution. Ultimately, rather than rewarding models that have "benchmaxxed" and produced potentially misleading safety scores, the field should reward transparent models that can reproducibly demonstrate their performance across a variety of benchmarks and shared user simulations.

Good infrastructure also necessitates closer interdisciplinary collaboration, with better incentive alignment towards a shared goal. This means setting aside resources ahead of time to work with lived-experience organizations, and prioritizing the sustainability of those relationships as existing benchmarks are maintained and updated. It would also benefit frontier AI labs to collaborate closely with academic institutions and set aside business incentives in favor of bringing intrinsic value to their users. Intrinsic value aside, regulation has already started to catch up and policy researchers are asking questions about the safety of their models, following the [recent deaths linked to AI chatbots]. [cite] Releasing some data to external researchers and getting more hands to work on improving model safety would be beneficial to all.

Lastly, one policy researcher who routinely works with legislators in the AI child safety space urged computer science researchers to prepare for testimony in court. In the absence of a corpus of human-LLM interaction data, the next best thing may be expert opinions. This way, lawmakers can make informed decisions without extensive documentation of bad outcomes, ideally preventing the accumulation of harms.

Conclusion

The main roadblock in mental-health AI evaluation is a shortage of the shared infrastructure needed to attribute meaning to the plethora of benchmarks in the field today. Misaligned incentives have created benchmarks that perform well in silos and restricted access to evidence, leaving policymakers with an unclear basis for regulation. At the same

time, mental health is an inherently pluralistic domain and these diverse perspectives on what constitutes harm, which populations are in need of safety guardrails and in what cultural contexts must be preserved. Ultimately, the infrastructure that is needed includes a technical comparative framework that leaves room for pluralism, and better collaborative networks that can facilitate more access to real-world data. The field needs to shift its attention towards building this infrastructure.

This matters most for the vulnerable users who are otherwise in harm's path in an unregulated space. To begin with, unsafe LLM interactions can have severe, even fatal consequences, and must be treated as a creditable dimension of AI safety. Despite risks, in places with poor mental healthcare resources, low affordability, or deep-rooted stigma, turning to an LLM may be the only viable option for people seeking help. Reliable safety evaluation tools for mental health cannot inherently eliminate the risks of ongoing human-LLM interactions, but they should give researchers, policymakers, and users a mode to identify what those risks are so that they can be mitigated. Without strong evaluation infrastructure, misleading safety scores risk blinding the field to the harms accumulating outside benchmark leaderboards, harms that affect people in real life.

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